INCH-POUND

MIL-DTL-10382E
8 JUNE 2026
SUPERSEDING
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6 October 1983

DETAIL SPECIFICATION

CORROSION PREVENTIVE, PETROLATUM, SPRAYING APPLICATION: FOR FOOD HANDLING MACHINERY AND EQUIPMENT

REACTIVATED AFTER 8 JUNE 2026 AND MAY BE USED FOR NEW AND EXISTING DESIGN AND ACQUISITIONS.

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of petroleum base corrosion preventive suitable for use by spraying application on food handling machinery and equipment.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory, Army Research Directorate, Specifications and Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-680 - Degreasing Solvent

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of this document are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

PPP-C-96 - Cans, Metal, 28 Gage and Lighter

PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)

PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 Through 12 Gallons)

FEDERAL STANDARDS

FED-STD-123 - Marking for Shipment (Civil Agencies)

FED-STD-791 - Testing Method of Lubricants, Liquid Fuels, and Related Products

(Copies of this document are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://ansi.org/.)

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ASTM INTERNATIONAL

ASTM-A109/A109M	-	Steel, Strip, Carbon, (0.25 Maximum Percent), Cold-Rolled
ASTM B117	-	Salt Spray (Fog) Apparatus, Operating
ASTM D93	-	Cup Tester, Closed, Flash Point by Pensky-Martens
ASTM D127	-	Standard Test Method for Drop Melting Point of Petroleum Wax, Including Petrolatum
ASTM D2500	-	Petroleum Products, Cloud Point of
ASTM D3951	-	Packaging, Commercial
ASTM D4057	-	Petroleum and Petroleum Products, Manual Sampling of

(Copies of these documents are available online at https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The corrosion preventive covered by this specification shall be a smooth, uniform mixture of such materials that the resulting compound shall conform to the requirements of this specification. The compound shall be dispersed in the solvent for ready application by spraying at room temperature and shall produce a thin, easily removable film upon drying (see 6,2). The material shall contain no compounds or any combination of materials, which might prove hazardous under the usages recommended by the manufacturer. The residual material after evaporation shall be harmless if inadvertently ingested by personnel.

3.2 Composition. The compound shall conform to the composition specified in table I.

TABLE I. Composition.

Ingredients	Percent by Weight
Petrolatum, white, U.S.P. Microcrystalline wax, white	31.0 ± 1.0
Lanolin, U.S.P.	10.5 ± 0.5
1-hydroxyethyl-2-heptadecenyl imidazoline	0.5 ± 0.05
2-amino-2-methyl-1-propanol	0.5 ± 0.05
Dry cleaning solvent MIL-PRF-680: Type IV or V	To make 100%

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3.3 Flash point. The minimum flash point of the corrosion preventive shall be 100°F (37.8°C) when tested as specified in 4.5.1.

3.4 Non-volatile matter. The corrosion preventive shall have a minimum non-volatile matter content of 40 percent, when tested as specified in 4.5.2.

3.5 Melting point. The non-volatile fraction shall have a minimum melting point of 150°F (65.6°C) when tested as specified in 4.5.3.

3.6 Abrasives. The corrosion preventive shall contain no abrasive substances. Tests shall be made as specified in 4.5.4.

3.7 Stability. The corrosion preventive shall show no separation or settling which is not easily reincorporated by mild agitation:

a. After being conditioned at 0° ± 3°F (-17° ± 1.7°C) for 7 days.
b. After being held at room temperature for an additional 21 days. Tests shall be made as specified in 4.5.5.

3.8 Ash. The compound shall not yield more than 0.01 percent ash upon ignition when tested as specified in 4.5.6.

3.9 Salt spray resistance. When tested for minimum of 7 days as specified in 4.5.7, there shall be no more than a trace of corrosion on any of the 3 test panels. Corrosion on the edges of the panels or on the surfaces within 0.25 inch from the edges shall be disregarded.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all corrosion preventive from the same batch or blending operation, subjected to the same processing operations and conditions and produced by one manufacturer and offered for delivery at one time.

4.3 Sampling.

4.3.1 For examination. Unless otherwise specified a random sample of filled containers shall be selected for examination in accordance with level I of ANSI/ASQ Z1.4. Additional guidance and sampling criteria may be referenced in MIL-STD-1916.

4.3.2 For tests. Prepare a 1 quart composite sample for tests in accordance with ASTM D4057. The sample shall be representative of the lot.

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4.4 Examination. Sample units selected in accordance with 4.3.1 shall be examined for visual defects from corrosion, and for defects in packaging, packing and marking for shipment. The defects and the acceptable quality levels shall be as shown in table II.

TABLE II. Classification of defects.

Item	AQL (percent)	Classification of Defects	Defect	Method of Inspection
Corrosion preventive (see 4.3.1 and 3.1)	2.5	Critical Major 101 Major 102	None defined Not a smooth and uniform mixture. Compound not. dispersed in the solvent.	 Visual Visual
Unit container (see 4.3.1 and 5.1)	2.5	Major 103 Major 104 Major 105 Major 106 Major 107	Improper type Improper size Improper fill [1/] Leakage Improper closure	Visual Visual Approved Scale [2/] Visual Visual
Box open (see 4.3.1 and 5.2)	2.5	Major 108 Major 109	Improper type [3/] Lack of, or improper strapping	Visual Visual
Box closed (see 4.3.1, 6.5.2, 6.5.3.1, or 6.5.3.2 and 6.5.3.3)	2.5	Major 110 Major 111 Major 112	Gross weight. max. Improper marking Improperly closed	Approved Scale [2/] Visual Visual

[1/] The actual weight: of a container filled with the minimum required quantities of corrosion preventive shall be the basis for determining the acceptable weight of subsequent: containers.
[2/] Approved by the procuring activity
[3/] When applicable

4.5 Tests.

4.5.1 Flash point. The flash point shall be determined as specified in ASTM D93.

4.5.2 Non-volatile matter. Two samples of the corrosion preventive shall be tested with a test being made on each sample as follows: Mix the composite sample (see 4.3.2) thoroughly by shaking. By means of a medicine dropper, transfer 1 to 2 g of the sample directly to a tared aluminum dish, and weigh to the nearest milligram. Transfer of the material and weighing should be made as rapidly as possible. Heat the weighed dish and contents in a mechanical convection oven at 105° to 110°C (221° to 230°F) for 3 hours. Cool in a desiccator for 30 minutes and weigh. From the weight of the residue left in the aluminum dish and the original weight of the sample taken, calculate the percent of non-volatile residue. Report the results as the average of the two tests.

4.5.3 Melting point. The melting point shall be determined as specified in ASTM D127.

4.5.4 Abrasives. Mix the composite sample (see 4.3.2) thoroughly by shaking. Mix approximately 75 ml of the composite sample with 200 ml of toluene, and stir until all of the soluble matter is in solution. Allow to stand for 1 hour at room temperature to permit any insoluble matter to settle. Carefully decant, wash the residue with 100 ml of toluene, and again carefully decant. Repeat the procedure with successively smaller portions of toluene until the solution is practically colorless. The residue after the last decantation shall be rubbed between two pieces of flat clean glass plate.

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The appearance of scratches on the glass plate shall be considered evidence of the presence of abrasive material.

4.5.5 Stability.

4.5.5.1 After 0 degrees F for 7 days. A glass jar shall be filled with the composite sample (see 4.3.2) to the level required by ASTM D 2500. The closed jar shall be held in a vertical position at 0° ± 3°F (-17° ± 1.7°C) for 168 hours. At the end of this period the jar and its contents shall be permitted to attain room temperature avoiding any physical disturbance of the corrosion preventive. The compound shall be considered as meeting this requirement if at the end of this period the corrosion preventive fluid regains homogeneity after mild agitation by slowly inverting the jar 4 times.

4.5.5.2 After being held at room temperature for an additional 21 days. The jar and its contents, tested in accordance with 4.5.5.1 for 168 hours, shall then be permitted to stand an additional 3 weeks (504 hours) at room temperature without physical disturbance. The compound shall be considered as having passed this test if, at the end of this period, the corrosion preventive compound regains its homogeneity after mild agitation by slowly inverting the tube 4 times.

4.5.6 Ash. Two samples of the corrosion preventive taken from the composite sample (see 4.3.2) shall be tested with a test being made on each sample as follows: Mix the composite sample thoroughly by shaking, and weigh 10 grams of the composite sample into a tared porcelain crucible and heat until the escaping fumes ignite. The sample shall be allowed to burn until the fire extinguishes itself, then the sample shall be further heated to drive off all volatile matter, and finally ignited over an open burner until free from carbonaceous matter. At no time shall the crucible be heated above a dull red heat. The crucible shall be cooled in a desiccator, weighed, and the percentage of ash calculated. Report the results as the average of the two tests.

4.5.7 Salt spray resistance.

4.5.7.1 Preparation of test panels. The test panels shall be approximately 2.0 by 3.0 by 0.25 inches, with well-rounded edges, having a 0.25 inch hole in one end for handling, and shall be prepared from open hearth low carbon steel conforming to ASTM A109. The steel shall be cold finished and dead soft in temper. The panel test surfaces shall be prepared by use of 280 grit silicon carbide or alundum abrasive paper so as to produce a surface finish of 10-20 microinches (R.M.S.). The surface shall be free of rust and shall be thoroughly cleaned using a method that will reduce all types of superficial contamination to the practical minimum. A cleaning method which has proved satisfactory is described in 6.3.

4.5.7.2 Treatment of panels. Three panels shall be coated in a room with the ambient temperature controlled at 77° ± 2°F (25°C) and relative humidity below 50 percent. Agitate the corrosion preventive compound (see 4.3.2) kept at this temperature in a manner which shall insure uniformity throughout, but which shall not whip air into the material. Apply the compound to 1 face of the panel by use of a doctor blade applicator so as to obtain a uniform coating 1 mil in thickness after drying. The coated specimen shall be permitted to dry in a horizontal position overnight. The edges of the panels shall then be protected by dipping them in molten microcrystalline wax or brushing molten microcrystalline wax on the 0.25 inch margins of the panel and cooling prior to test. Care shall be taken that the film of compound on the panel is not disturbed.

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4.5.7.2 Procedure. Three test panels prepared in accordance with 4.5.7.1 shall be exposed for 7 days of salt spray in accordance with ASTM B117. The test panels shall be located with the holes downward and shall be held at an angle of 15 degrees from the vertical and shall be parallel to the principal direction of horizontal flow of fog through the chamber. The compound shall be considered as having passed this test if, at the end of the test period, none of the panels shows more than a trace of corrosion. A trace of corrosion shall be defined as the presence of not more than 3 spots of corrosion, no one of which is larger than l mm in diameter. Only that corrosion within the significant areas which includes all but a 0.25 inch margin of the panels shall be considered.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.1.1 Packaging and packing notes. Due to the substantial passage of time since the last revision of this specification, additional packaging information that may be helpful including past definitions and procedures for unit of product, sampling, packaging inspection, and exterior container markings are included in 6.5 and its sections therein. This legacy content will be deleted in the next revision or amendment to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This material is intended as a preservative for food preparation machinery and equipment and is intended to prevent corrosion of the culinary equipment and other materials that contact food while not introducing residual toxic materials or residuals that could remain after removal (see 6.3) before use with food and be hazardous when ingested.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Quantity desired.
c. Size of container required (see 5.1).
d. Required levels of packaging (see 5.1) and packing (see 5.2).
e. Special marking, if required (see 5.3.1 or 5.3.2).

6.3 Removability. This corrosion preventive is easily removed with hot water 160°F (71°C).

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6.4 A method of specimen preparation for protection tests. The following procedure has been found to produce duplicable specimen surfaces of a high level of cleanliness:

a. After rounding the edges of the panel, reaming out the holes used for suspension, etc., wipe the surfaces as clean as possible by use of solvent-soaked rags.
b. Scrub the panel with a clean surgical gauze swab in hot petroleum naphtha (VM&P grade).
c. Rinse in hot petroleum naphtha.
d. Rinse in hot commercial anhydrous methyl alcohol. Store in desiccator until used.
e. Buff the panels, ending with a new section of 280 grit carborundum or alundum abrasive paper.
f. Wipe off all superficial dusts from the abrasion operation, using clean cloths or absorbent tissue.
g. Scrub abraded face of panel thoroughly with a dry surgical gauze pad held in a blotter-type holder, until there is no dark stain on a clean section of the gauze.
h. Spray the panel with clean naphtha using a paint spray gun. The panel should be held in a rack at 25 degrees from the vertical. The spray should be directed vertically downward on the panel flushing the test surface progressively downward. Spray the test surface, back face, and finally the test surface again.
i. Rinse in hot naphtha.
j. Rinse in hot commercial anhydrous methyl alcohol.

6.5 Packaging and packing considerations. The previous packaging and packing requirements are retained for guidance purposes only. Procurement actions should follow standard packing procedures in 5.1.

6.5.1 Packaging. Packaging should be level A, B, or Commercial as specified (see 6.2).

6.5.1.1 Level A. The corrosion preventive should be delivered in 1 gallon cans, 5 gallon steel pails, or 55 gallon steel drums, as specified (see 6.1). One gallon cans should conform to type V, class 2 round of FPP-C-96. Each can should be exterior coated in accordance with Plan B of PPP-C-96. Five-gallon steel pails should conform to type II of PPP-P-704. Fifty-five gallon drums should conform to type I of PPP-D-729.

6.5.1.2 Level B. The corrosion preventive should be packaged as specified in 6.5.1.1 except that the exterior coating of the 1 gallon cans should conform to Plan A of PPP-C-96.

6.5.1.3 Commercial. The corrosion preventive should be packaged in accordance with supplier's commercial practice. Protection should be such as to prevent deterioration during shipment and ensure safe delivery at destination.

6.5.2 Packing. Packing should be level A, B, or Commercial as specified (see 6.2).

6.5.2.1 Level A. The 1 gallon containers should be packed in accordance with the overseas shipment requirements of the Appendix to PPP-C-96. The 5-gallon and 55 gallon containers should require no over-packing.

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6.5.2.2 Level B. The 1 gallon containers should be packed in accordance with the domestic shipping requirements specified in the Appendix to PPP-C-96. The 5 gallon and 55 gallon containers should require no overpacking.

6.5.2.3 Commercial. The corrosion preventative should be packed in accordance with ASTM D3951.

6.5.3 Marking.

6.5.3.1 Civil agencies. In addition to markings required by the contract or order, and 6.5.3.3 unit containers, and shipping containers should be marked in accordance with FED-STD-123.

6.5.3.2 Military agencies. In addition to markings required by the contract or order, and 6.5.3.3, unit containers, and shipping containers should be marked in accordance with MIL-STD-129.

6.5.3.3 Labels. In addition to the marking specified in 6.5.3.1 or 6.5.3.2 (as applicable) each unit container should be labeled with a caution as follows:

"CAUTION:

a. Do not use for cooking or baking.
b. Use only for corrosion prevention during storage.
c. Remove the preventive from food handling equipment and machinery before operating by thoroughly flushing with hot water (160°F)."

6.6 Metric units. When metric dimensions are required, units for inch, foot, foot-pounds and feet per second may be converted to the metric equivalent by multiplying them by the following conversion factors listed in table III.

TABLE III. Conversion factors. 1/

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/sec	0.3048	=	meter per second (m/s)
Fahrenheit (°F)	$\frac{t°F - 32}{1.8}$	= t°C	Celsius (°C)

1/ Conversion factors can be associated with ASTM E380 entitled “Metric Practice Guide.”

6.7 Subject term (key word) listing.

Culinary Preservation Storage

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - MR
Navy - SH

Preparing activity:
ARMY - MR

(Project 8030-2026-013)

Review activities:
Army - GL
Navy - AS, CG

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